

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY****Second Semester of B. Tech. (CE/IT/EC/CSE) Theory Examination****April-May 2018****ME144 Elements of Engineering****Date: 08/05/2018, Tuesday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I must be attempted in drawing sheets and Section II must be attempted in answer sheets.
3. Make suitable assumptions and draw neat figures wherever is required.
4. Use of scientific calculator is allowed.
5. Figures drawn in the question paper are not to the scale.
6. Dimensions shown in the figure are in mm.
7. Show all the necessary dimensions in the solution.
8. Figures to right indicate the full marks.

**SECTION – I****Q - 1 Do as directed.**

1. Justify the statement, “Engineering Graphics is the universal language and tool of communication for engineers”. [01]
2. 1 decimeter = \_\_\_\_\_centimeters. [01]  
a) 10                      b) 100                      c) 1000                      d) 10000
3. In which type of following dimensioning system, the dimension lines are broken near the middle for inserting the dimensions. [01]  
a) Aligned                      b) Unidirectional                      c) Inline System                      d) None of these  
System                      System
4. Why second and fourth quadrants are not used in practice? [01]
5. The isometric drawing of a circle is a(n) \_\_\_\_\_. [01]  
a) Ellipse                      b) Parabola                      c) Hyperbola                      d) Circle
6. When a solid is cut by a plane parallel to its base, thus removing the top portion, the remaining portion is called its frustum. True or False (T/F)? [01]
7. Inscribe the names of CAD Modeling software. [01]

**Q – 2.a** Two fixed straight lines OX and OY are at right angle to each other. A point S is at a distance of 45 mm from OX and 55 mm from OY. Draw a rectangular hyperbola passing through point S. Draw normal and tangent at any point on the curve. [06]

**Q – 2.b** A cone has 40 mm diameter of base and 50 mm height of axis, is resting on one of the generator on HP. The plan of axis makes  $30^\circ$  inclination with VP. Draw the projections of the cone. [06]

**OR**

**Q – 2.b** A cylinder has 50 mm diameter of base and 72 mm height of axis, is resting on the H.P. on its base. It is cut by an A.I.P. in such a way that the true shape of the section is an ellipse of largest possible major axis. Draw the sectional plan and true shape of section. Find the inclination of the sectional plane with HP. [06]

**Q – 2.c** Discuss the importance of CAD software. [04]

- Q – 3.a** Figure 1 shows a pictorial view of an object. Draw the following views: (1) Front view, [06]  
(2) Top view. Dimensions shown in the figure are in mm.

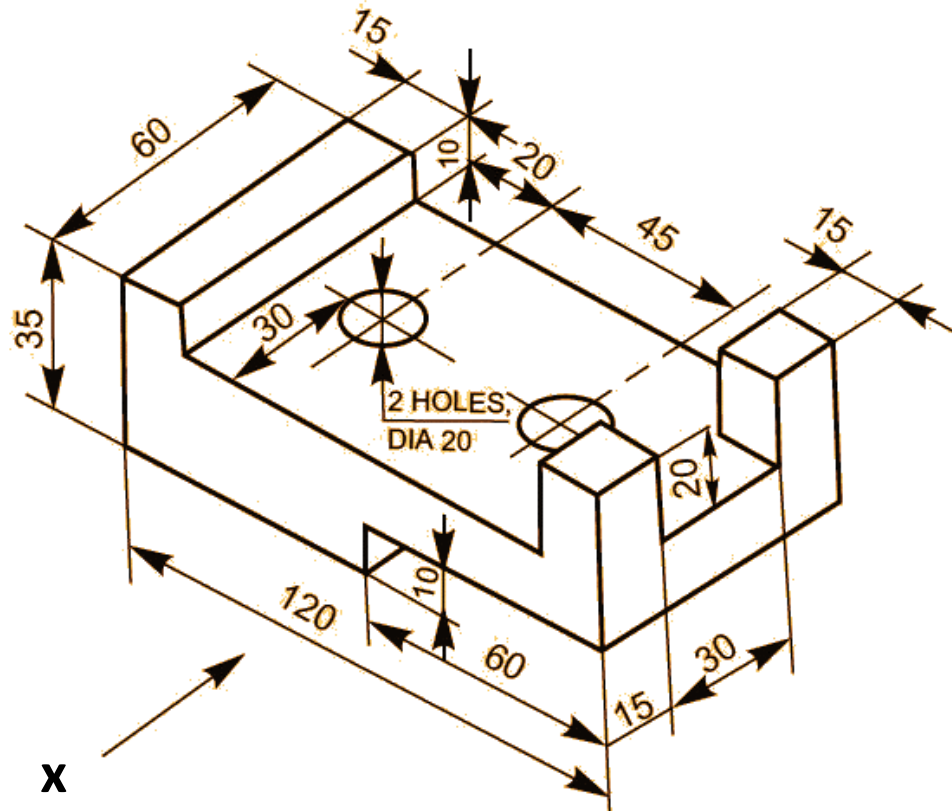


Figure 1

- Q – 3.b** A pentagonal prism has a side of base 40 mm and height 100 mm long, is resting on its [06]  
base on HP. Draw the isometric drawing of the pentagonal prism.

OR

- Q – 3.b** Figure 2 shows the orthographic projections of an object in first angle projection system. [06]  
Draw its isometric view. Dimensions shown in the figure are in mm.

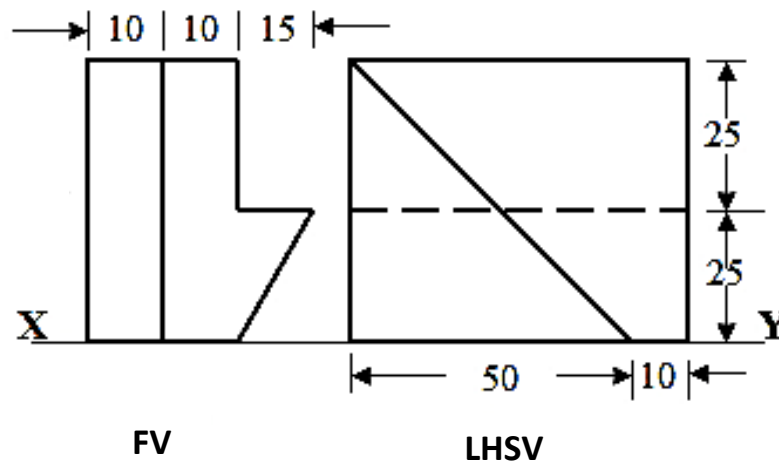


Figure 2

**SECTION – II****Q - 4 Choose one correct alternative of the following multiple-choice questions.**

1. The main role of civil engineer is \_\_\_\_\_. [01]  
 a) To maintain & repair      b) To invite tenders      c) Planning, design & construction of structures      d) All of these
2. Which branch of civil engineering deals with planning, designing and construction of roads, railways, ports and airports? [01]  
 a) Construction Planning & Management      b) Transportation Engineering      c) Geotechnical Engineering      d) Structural Engineering
3. In which type of building, the thickness of wall remains same throughout the whole building \_\_\_\_\_. [01]  
 a) Load bearing structure      b) Framed structure      c) Composite structure      d) None of these
4. The main principle of surveying is \_\_\_\_\_. [01]  
 a) Work from part to the whole      b) Higher to lower level      c) Work from whole to the part      d) Lower to higher level
5. The residential building with three-side margin is called as \_\_\_\_\_. [01]  
 a) Detached house      b) Semi-detached house      c) Row house      d) Pent house
6. What is the ideal length to width ratio as per principle of planning? [01]  
 a) 1.2 to 1.5      b) 1 to 1.2      c) 1.5 to 2      d) More than 2
7. What should be the width of footing, if the thickness of wall is 30 cm? [01]  
 a) 110 cm      b) 120 cm      c) 70 cm      d) 90 cm

**Q – 5.a** A railway map showing 40 m distance as 1 cm long line. Construct a diagonal scale to read a single meter and maximum distance up to 600 m on it. Find its R.F. and show 438 m on it. [06]

**Q – 5.b** Enlist the name of building components. Explain the function of plinth and foundation. [04]

**OR**

**Q – 5.b** List out the principle of building planning and explain grouping in detail. [04]

**Q – 5.c** Define boiler. Enlist the essential parts of boiler. [02]

**Q – 5.d** Prove that  $c_p - c_v = R$ . [03]

**Q – 6.a** Compare the following terms. (Attempt any two)

(i) Wet steam v/s Dry saturated steam [02]

(ii) Fire tube boiler v/s Water tube boiler [02]

(iii) Natural circulation boiler v/s Forced circulation boiler [02]

**Q – 6.b** Explain any two strokes of 4-stroke Spark Ignition (SI) engine cycle with neat sketches. [04]

**OR**

**Q – 6.b** In the context of internal combustion engine, define the following terms. [04]

(i) Clearance volume (ii) Stroke (iii) Compression ratio (iv) Flywheel

**Q – 6.c** In a cyclic process, four heat transfers as,  $Q_{1-2} = +925 \text{ kJ}$ ,  $Q_{2-3} = -110 \text{ kJ}$ ,  $Q_{3-4} = -770 \text{ kJ}$ ,  $Q_{4-1} = +220 \text{ kJ}$ . The work done during three processes are  $W_{1-2} = +70 \text{ kJ}$ ,  $W_{2-3} = -50 \text{ kJ}$ , and  $W_{4-1} = +170 \text{ kJ}$ . Calculate the work done during process 3-4. What is the change in internal energy during cyclic process? **[02]**

**Q – 6.d** Explain domestic vapor compression refrigerator with a neat sketch. **[03]**

**OR**

**Q – 6.d** Explain window air conditioner system with a neat sketch. **[03]**

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